

Experiment: 3

1. Important concepts of React. js

- a. Write a React program to implement a counter button using React use State hook

SOURCE CODE:

The below code save as App.js:

```
import React, { useState } from 'react';

const App = () => {
  const [count, setCount] = useState(0);

  const handleClick = () => {
    setCount(prev => prev + 1);
  };

  return (
    <div style={{ textAlign: "center", marginTop: "50px" }}>
      <h2>Count: {count}</h2>
      <button onClick={handleClick}>Increment</button>
    </div>
  );
};

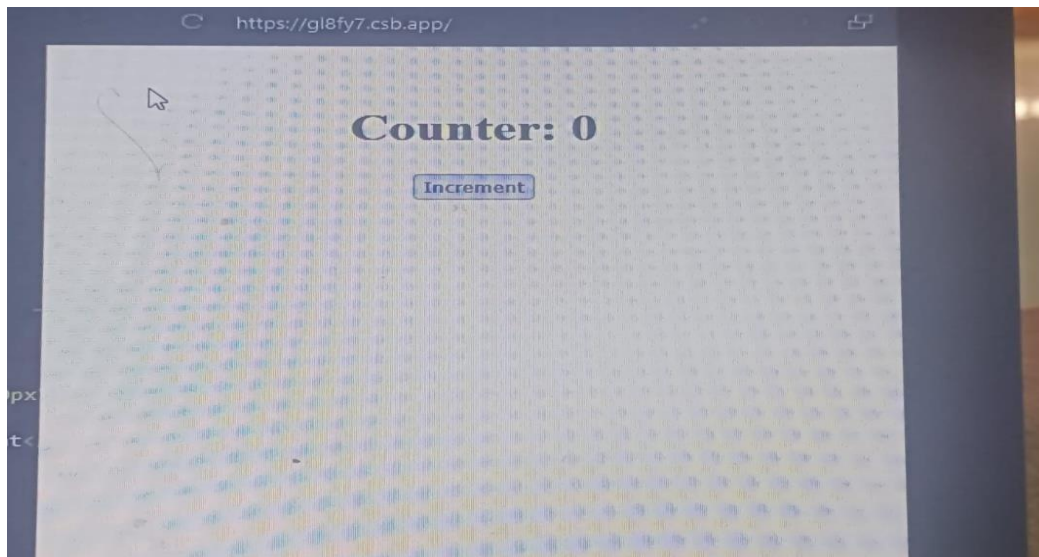
export default App;
```

The below code save as index.js:

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(<App />);
```

OUTPUT:



b. Write a React program to fetch the data from an API using React use Effect hook

SOURCE CODE:

The below code save as App.js:

```
import React, { useEffect, useState } from 'react';

const App = () => {
  const [users, setUsers] = useState([]);

  // Fetch users on component mount
  useEffect(() => {
    fetch('https://jsonplaceholder.typicode.com/users')
      .then(response => response.json())
      .then(data => setUsers(data))
      .catch(error => console.log("Error fetching data:", error));
  }, []); // empty dependency array means run once on mount

  return (
    <div style={{ padding: '20px' }}>
      <h2>User List</h2>
      {users.length === 0 ? (
        <p>Loading...</p>
      ) : (
        <ul>
          {users.map(user => (
            <li key={user.id}>
              <strong>{user.name}</strong> - {user.email}
            </li>
          ))}
        </ul>
      )}
    </div>
  );
};

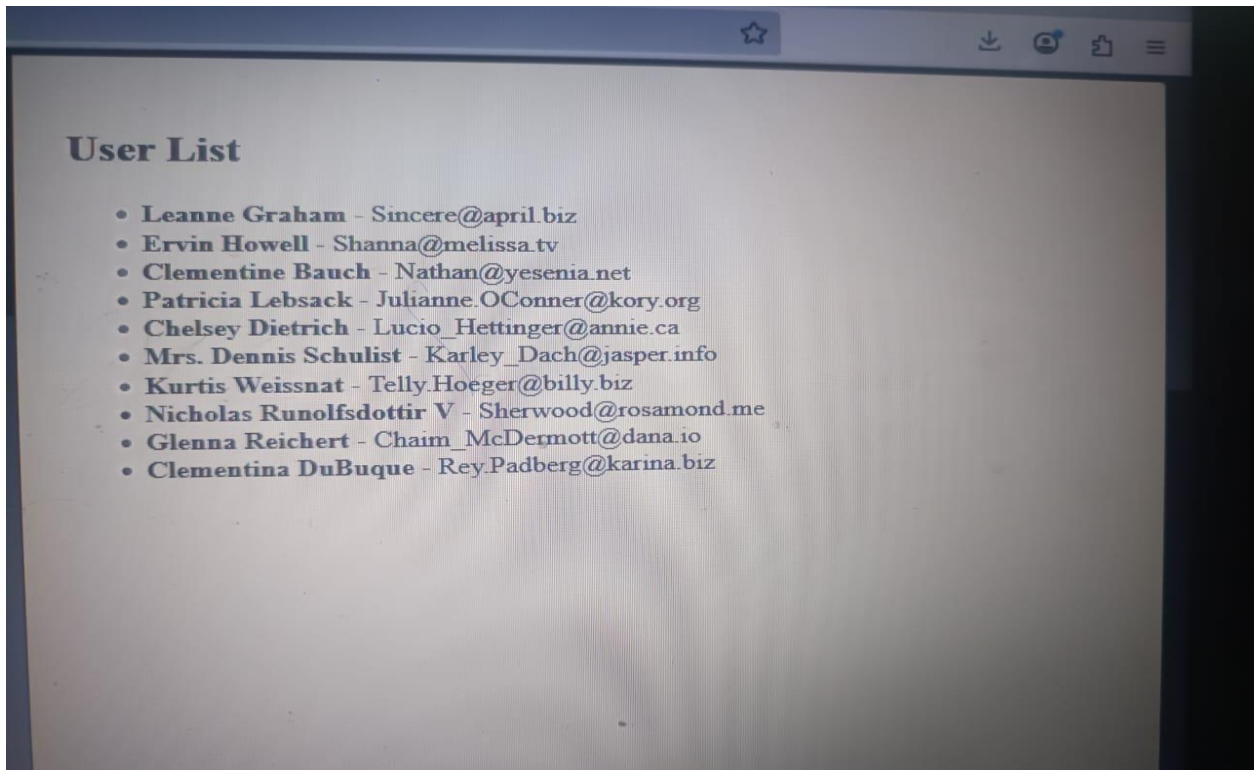
export default App;
```

The below code save as index.js:

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(<App />);
```

OUTPUT:



- c. Write a React program with two react components sharing data using Props.

SOURCE CODE:

The below code as App.js:

```
import React from 'react';
import Greeting from './Greeting';
import Info from './Info';

const App = () => {
  const user = {
    name: 'Inbha',
    age: 13,
    hobby: 'Exploring Space'
  };

  return (
    <div style={{ textAlign: 'center', marginTop: '50px' }}>
      <Greeting name={user.name} />
      <Info age={user.age} hobby={user.hobby} />
    </div>
  );
};

export default App;
```

The below code as Greeting.js:

```
import React from 'react';

const Greeting = ({ name }) => {
  return <h2>Hello, {name}! 🖐️</h2>;
};

export default Greeting;
```

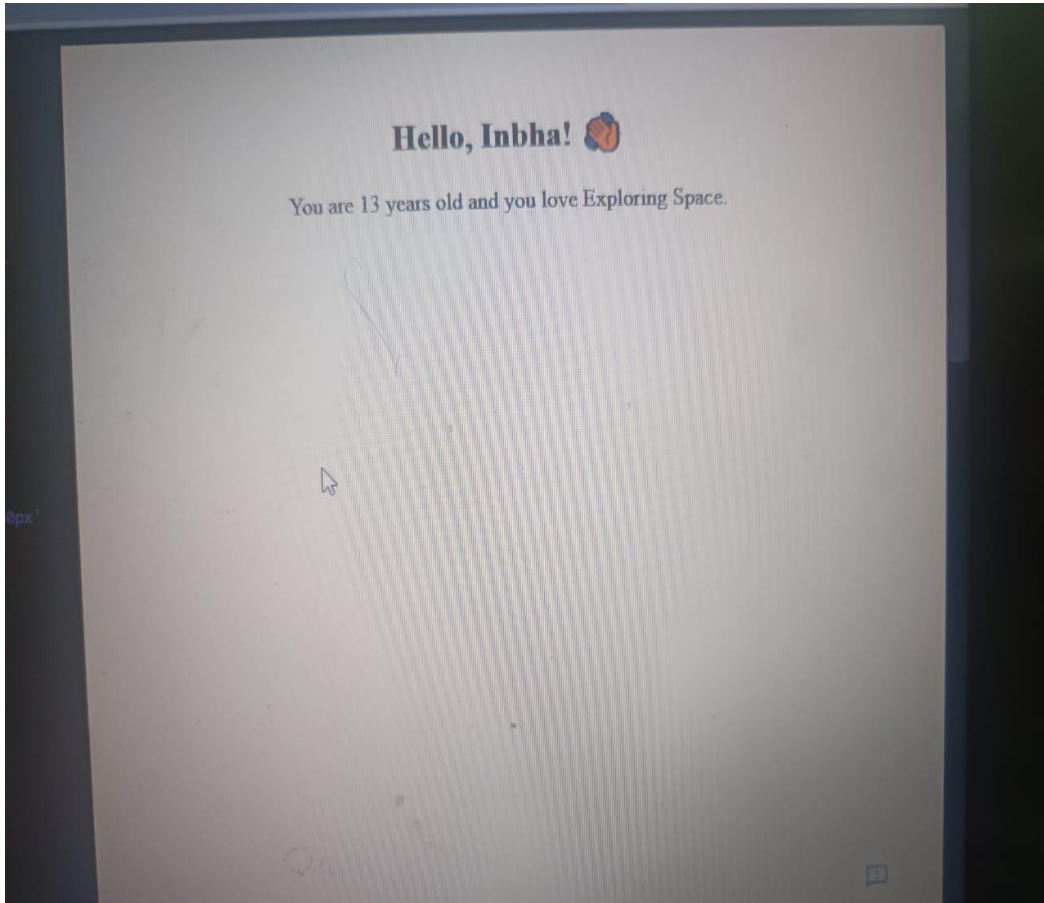
The below code save as Info.js:

```
import React from 'react';

const Info = ({ age, hobby }) => {
  return (
    <p>
      You are {age} years old and you love {hobby}.
    </p>
  );
};
```

export default Info;

OUTPUT:



d. Write a React program to implement the forms in react

SOURCE CODE:

The below code save as App.js:

```
import React, { useState } from 'react';

const App = () => {
  const [formData, setFormData] = useState({
    name: "",
    email: ""
  });

  const [submittedData, setSubmittedData] = useState(null);

  const handleChange = (e) => {
    setFormData({
      ...formData,
      [e.target.name]: e.target.value
    });
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    setSubmittedData(formData);
    setFormData({ name: "", email: "" });
  };

  return (
    <div style={{ maxWidth: '400px', margin: '50px auto', textAlign: 'center' }}>
      <h2>User Form</h2>
      <form onSubmit={handleSubmit}>
        <div>
          <input
            type="text"
            name="name"
            placeholder="Enter your name"
            value={formData.name}
            onChange={handleChange}
            required
          />
        </div>
        <br />
        <div>
          <input
            type="email"
            name="email"
            placeholder="Enter your email"
            value={formData.email}
          />
        </div>
      </form>
    </div>
  );
};
```

```

      onChange={handleChange}
      required
    />
  </div>
  <br />
  <button type="submit">Submit</button>
</form>

{submittedData && (
  <div style={{ marginTop: '30px' }}>
    <h3>Submitted Data:</h3>
    <p>Name: {submittedData.name}</p>
    <p>Email: {submittedData.email}</p>
  </div>
)}
</div>
);
};

```

export default App;

The below code save as index.js:

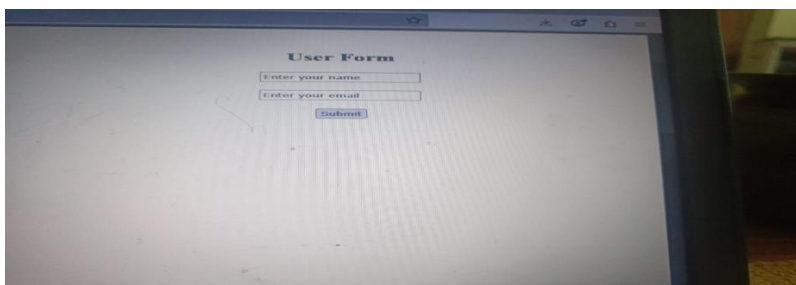
```

import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(<App />);

```

OUTPUT:



User Form

Enter your name

Enter your email

Submit

Submitted Data:

Name: P ROHINI BAI

Email: rohini.121987@gmail.com

- e. Write a React program to implement the iterative rendering using map() function.

SOURCE CODE:

The below code save as App.js:

```
import React from 'react';

const App = () => {
  const fruits = ['Apple', 'Banana', 'Cherry', 'Mango', 'Orange'];

  return (
    <div style={{ textAlign: 'center', marginTop: '50px' }}>
      <h2>Fruit List</h2>
      <ul style={{ listStyle: 'none' }}>
        {fruits.map((fruit, index) => (
          <li key={index} style={{ padding: '5px' }}>
            🍌 {fruit}
          </li>
        ))}
      </ul>
    </div>
  );
};

export default App;
```

The below code save as index.js:

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(<App />);
```

OUTPUT:

